

Galois Fields of type $GF(p^n)$

For encryption algorithms that operate on 2, 3, 4, ..., n bits of data at a time we require finite fields of order $2^2, 2^3, 2^4, \dots, 2^n$.

The encryption algorithms used in the Advanced Encryption Standard (AES) operate on 8-bits of data at a time and require a finite field of order 256. This is accomplished with 256 polynomial elements that are multiplied and added modulo $m(x)$, an irreducible polynomial of degree 8.

Example: Describe a finite field that can be used for 3-bit encryption algorithms.

For 3-bits we need a finite field of order 8.

Note that $\{\mathbb{Z}_8, +, \times\}$ is not a finite field.

Instead we use polynomial elements $\{0, 1, x, x+1, x^2, x^2+1, x^2+x, x^2+x+1\}$ with multiplication and addition modulo $m(x) = x^3 + x + 1$.

Addition $GF(2^3)$

+	000	001	010	011	100	101	110	111
000	000							
001	001	000						
010	010	011	000					
011	011	010	001	000				
100	100	101	110	111	000			
101	101	100	111	110	001	000		
110	110	111	100	101	010	011	000	
111	111	110	101	100	011	010	001	000

Multiplication $GF(2^3)$

.	000	001	010	011	100	101	110	111
000	000							
001	000	001						
010	000	010	100					
011	000	011	110	101				
100	000	100	011	111	110			
101	000	101	001	100	010	111		
110	000	110	111	001	101	011	010	
111	000	111	101	010	001	110	100	011

Exercises:

1. Show how each of the **three** representations shown in the $GF(2^3)$ addition table are derived by:
 - Converting the associated row/column into polynomials and adding
 - Reducing the result mod 2 and converting back to binary

+	000	001	010	011	100	101	110	111
000	000							
001	001	000						
010	010	011	000					
011	011	010	001	000				
100	100	101	110	111	000			
101	101	100	111	110	001	000		
110	110	111	100	101	010	011	000	
111	111	110	101	100	011	010	001	000

2. Show how each of the **three** representations shown in the $GF(2^3)$ multiplication table are derived by:
 - Converting the associated row/column into polynomials and multiplying
 - Reducing the result mod $m(x) = x^3 + x + 1$ and converting back to binary

.	000	001	010	011	100	101	110	111
000	000							
001	000	001						
010	000	010	100					
011	000	011	110	101				
100	000	100	011	111	110			
101	000	101	001	100	010	111		
110	000	110	111	001	101	011	010	
111	000	111	101	010	001	110	100	011

3. Show that for $GF(2^3)$ addition:
 - $(x^2 + x) + (x^2 + 1)$ can be computed using the bitwise XOR operation: $(110) \oplus (101)$
 - $(x^2 + x + 1) + (x^2 + x)$ can be computed using the bitwise XOR operation: $(111) \oplus (110)$
4. Show that for $GF(2^3)$ multiplication:
 - $(x) \cdot (x)$ can be computed using a 1-bit left shift operation: b_1b_00 on (010)
 - $(x) \cdot (x^2 + x)$ can be computed using a 1-bit left shift operation: b_1b_00 on (110) and then evaluating $(b_1b_00) \oplus (011)$.

5. In $GF(3^2)$ there are 9 polynomial elements $\{0, 1, 2, x, x+1, x+2, 2x, 2x+1, 2x+2\}$ with multiplication and addition modulo $m(x) = x^2 + 1$. In $GF(3^2)$ evaluate:

- $(02) + (21)$
- $(12) + (22)$
- $(02) \times (12)$
- $(20) \times (10)$

Answers... for last question

In $GF(3^2)$ there are 9 polynomial elements $\{0, 1, 2, x, x+1, x+2, 2x, 2x+1, 2x+2\}$

with multiplication and addition modulo $m(x) = x^2 + 1$. In $GF(3^2)$ evaluate:

- $(02) + (21)$ **20**
- $(12) + (22)$ **01**
- $(02) \times (12)$ **21**
- $(20) \times (10)$ **01**